SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.*		
Student Name:	J. MEHER KOUSHIK	**Reg. No.:**	192525211
Section / Batch:	AI&ML	**Date:**	31/08/2026

Code of Conduct

I J. MEHER KOUSHIK (1925252111) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: J. MEHER KOUSHIK

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 **Minimum Spanning Tree - Kruskal's Algorithm** 20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected

Your Answer

```
KRUSKAL(G)
1. MST ← empty set
2. Sort all edges of G in increasing order of cost
3. For each vertex v in G:
      MAKE-SET(v)
4. For each edge (u, v) in sorted edges:
```

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Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q2 **Graph - Topological Sort** 20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach

Your Answer

```
TOPOLOGICAL-SORT(G)

1. For each vertex v in G:
      state[v] ← UNVISITED

2. Stack ← empty
```

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Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

Q3 **Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
SMART-TRANSPORT(G, Source, Destination)
1. MST ← KRUSKAL(G)
2. Create graph T using MST edges
3. Set distance[Source] = 0
   Set all other distances = ∞
4. While unvisited vertices exist:
```

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Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
SHORTEST-PATH(G, S, D)
1. Set dist[v][0] = ∞ and dist[v][1] = ∞
2. dist[S][0] = 0
3. Put (S, 0) in PriorityQueue
4. While PriorityQueue is not empty:
     (u, used) ← Extract-Min()
```

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Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q5 **Minimum Spanning Tree - Prim's or Kruskal's Algorithm** 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
SECOND-BEST-MST(G)
1. MST ← KRUSKAL(G)
2. Best ← weight(MST)
3. Second ← ∞
4. For each edge e in MST:
     Remove e from G
```

Save Answer

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My Assessments

Course: **CSA0305RA**

PSEUDOCODE WRITING TASK OPEN ✓ Completed
CO5_AT1_Pseudocode Writing Task
5/5 answered
Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed
DEBUGGING & OPTIMIZATION TASK
10/10 answered
Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed
DEBUGGING & OPTIMIZATION TASK
10/10 answered
Review →

CO3_AT3_MCQ OPEN ✓ Completed
Multiple Choice Questions - Non Linear Data Structures
25/25 answered
Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed
C CODE OUTPUT PREDICTION
Review →